MSc Group 2 Project

ClearCare

Week 3 update

A record of

What inspectors found.

What the provider planned.

Whether the next inspection **found it fixed.**

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Week 3 update

Reflections

Last week's feedback, in four topics.

1 **Qualitative research** — Talk to real people about real problems — Slides 3–4

2 **Technologies** — The choice of backend and hosting — Slide 5

3 **Data visualisation** — Show a lot of inspection data effectively — Slide 6

4 **Artificial intelligence** — Why build this when anyone can ask a chatbot — Slide 7

1 Feedback 1 of 4

Qualitative research

This week

We need real people:

two interviews this week.

Today

Availability over preference?

Ahead: demand will grow faster than supply

+162%

population aged 85+ by 2045

Department of Health projections, via Nursing Homes Ireland

1,000

extra nursing-home beds a year, to 2040

ESRI

95%

occupancy already, in nursing homes

Nursing Homes Ireland

1 Feedback 1 of 4

Interviews: how many families read the reports

Director of Nursing

Private nursing home, Dublin · 29 Sept

> "Not everyone checks the reports, maybe **one out of five or ten.**

Assistant Director of Nursing

Private nursing home, Dublin · 30 Sept

> "**All families** check reports."

Two interviews, 29 and 30 Sept, with signed consent.

Technologies

Decision 1 · Hosting

Hosting on Oracle Cloud (OCI)

One provider for the API, the pipeline, the database and the PDF archive.

Decision 2 · Backend

FastAPI → Spring Boot

- Mature ecosystem, supported for the life of the project.

How we judge “enterprise ready”

Evaluation

gold set, measured accuracy

Security

Spring Security

Monitoring

health checks, job alerts

Testing

real PostgreSQL in CI

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Data visualisation

Goal: show the most information with the least display.

Compliance per regulation over time
Bar charts, with the legend shown at right

A mind map of each report
Regulations, judgments, the provider's promises

Maximum agreed price per home
From the HSE

Activities and services
In the inspector's words



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AI vs ClearCare

General AI chatbot	ClearCare
You find and upload each report	Every HIQA report collected automatically and kept current
Answers can vary between runs	Judgments extracted once, stored, the same for every user
Paraphrases loosely, may hallucinate	Every line linked to the inspector's words and the PDF page
Compares anything and may sound confident	Says “not comparable” when inspections covered different regulations
Conversation only	One page per home: findings, promises, what the next inspection said
No memory across reports	Compliance history over time

Week 3 update

We will listen to the advice given.

Next: the retrospective — what went well, what did not, what comes next

Sprint review

Retrospective I

What went well

- Conducted two interviews; emailed Sage Advocacy.
- Defined the coding standards and conventions.
- Deployed the first Next.js app.
- Re-evaluated the technologies.
- Audited a few HIQA reports.
- Had the first meeting with our mentor.

What went poorly

- Sage Advocacy has not replied yet.
- Less coding than we wanted.
- The Spring Boot backend is not started yet.

Sprint review

Retrospective II

New ideas

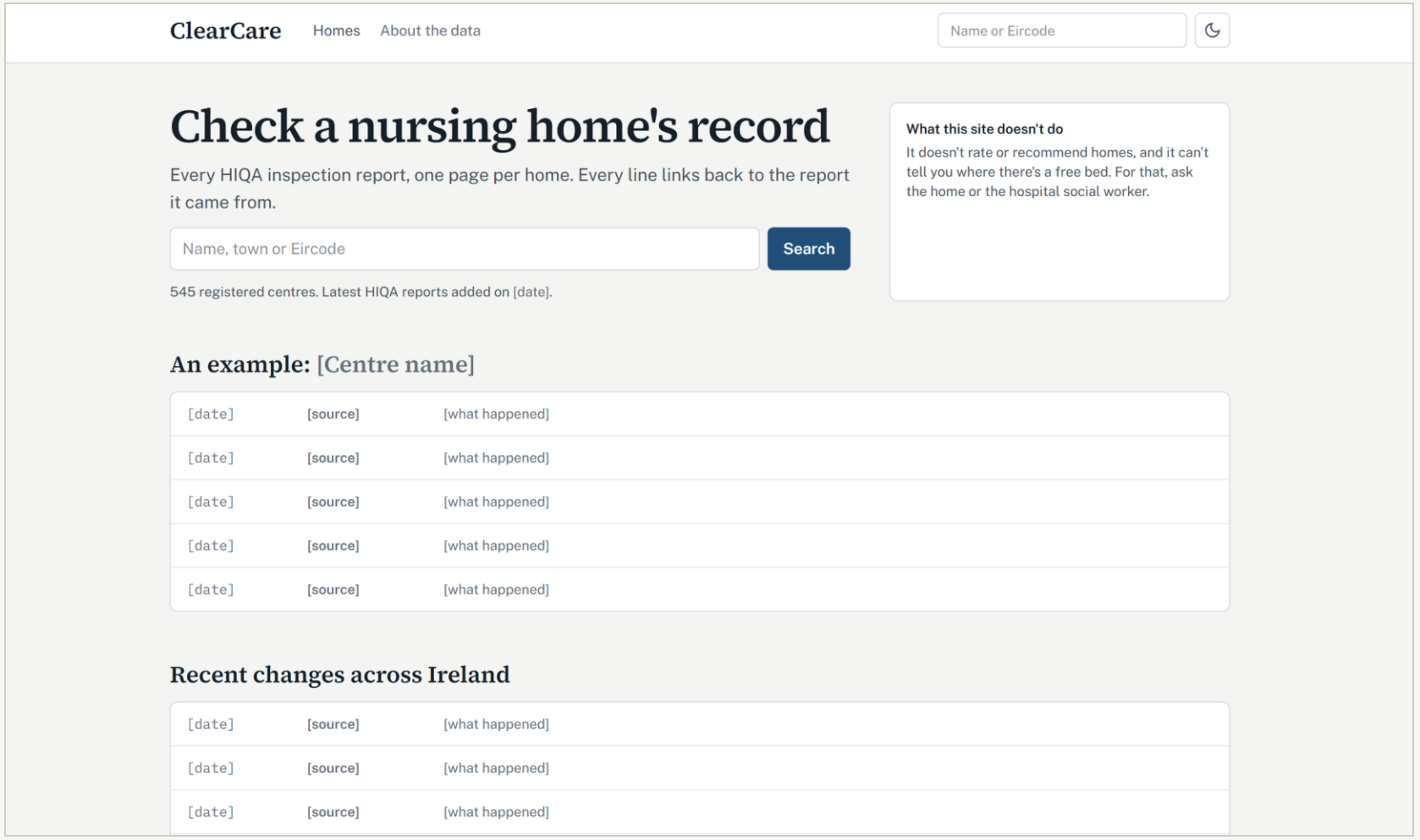
1. **Show where homes are**
   On a map
2. **Add the financial side**
   Maximum contract price
3. **Mobile-friendly UI**

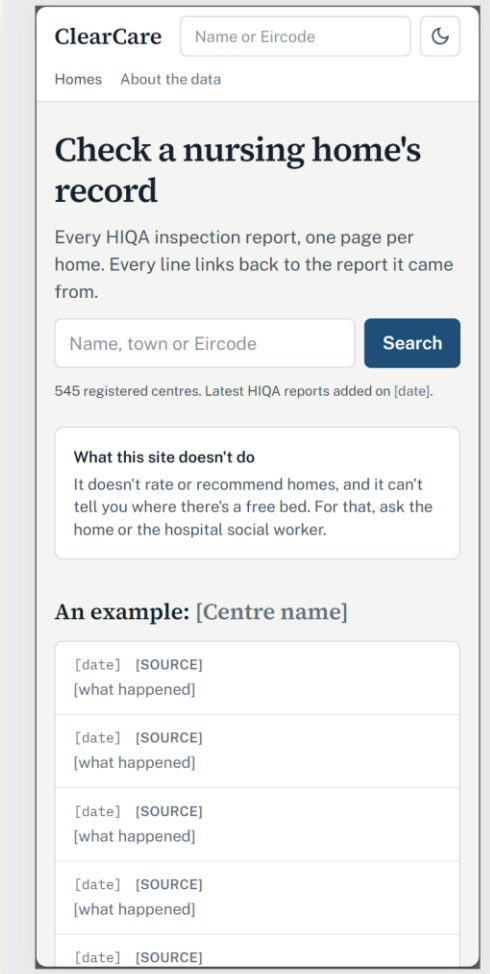
Actions for next week

1. Further outreach for interviews
2. Start Spring Boot backend
3. More coding!

Vercel deployment

clear-care-nine.vercel.app





Requirements specification

Req ID	Name of Req	Description	Priority	Who uses it?	Release/Week
1	Ingest HIQA register	Ingest the HIQA register of designated nursing centres (CSV) to build the master list of homes and providers.	High	System	4
2	Geocode home addresses	Geocode home addresses to coordinates (LocationIQ), which enables radius search.	Medium	System	5
3	PDF ingestion and archiving	Ingest and archive each HIQA PDF as the source of truth, and keep a record of it in the database.	High	System	4
4	Report metadata and amenities extraction	Extract inspection metadata (first page of each report) and any amenities mentioned, to drive the listing page and the filters.	High	System	4
5	Section-aware report parsing	Parse each report into sections, keeping page references and section metadata.	High	System	4
6	Regulation-level judgement extraction	substantially compliant / not compliant) with a confidence score, plus the inspector's supporting text and page.	High	System	5
7	Scheduled report crawler	Setup and run a scheduled crawler that finds and ingests new or updated HIQA reports.	Medium	System	6
8	User registration and login	Users can register and log in.	High	User	6
9	Browse general listing (landing page)	Users can browse the general listing of homes without an account.	High	User	4
10	Search nursing home by name	Users can search for a home by name.	High	User	4
11	Filter by location	Users can filter homes by location (county, radius).	High	User	5
12	Filter by amenities and preferences	Users can filter by amenities, services and other preferences, combinable with the location filter.	Medium	User	5
13	Compliance history over time	Show each home's compliance history over time, not just the latest report.	High	User	6
14	Shortlist management	Users can add, remove and view a shortlist of homes.	Medium	User	6
15	Side-by-side grid comparison	Users can compare shortlisted homes in a side-by-side grid.	Medium	User	7
16	Flag non-comparable judgements	Tell the user when judgements for two homes cannot be fairly compared.	Medium	User	8
17	Show source evidence for judg	Show the inspector's own words and link to the source PDF page behind every judgement.	Medium	User	8
18	Semantic search	Semantic search over report text (embeddings in pgvector).	Low	User	9
19	Compliance trend alerts	Compliance trend alerts for shortlisted/tracked homes.	Low	User	9